BASE CHANGE FOR A PUSHFORWARD OF A SHEAF OF SETS OVER THE SPECTRUM OF A SEPARABLY CLOSED FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the localisation and the cofiltered limit presentation, the reduction to finite type, and the application of the general base change lemma with the geometric connectedness input.

Editorial note. The label lemma-base-change-f-star-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's cartesian diagram is the author's and is delivered verbatim; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Base change for a filtered colimit of schemes. Same chapter, the lemma comparing Rg_* of a colimit with the colimit of the comparisons, used to reduce to finite type over S. Supplies the passage from an affine X presented as a cofiltered limit to a finite type model.
- Base change in the presence of geometrically connected fibres. Same chapter, the general base change lemma for a flat morphism whose strict henselisations have geometrically connected fibres. Supplies the conclusion once the fibres are known to be geometrically connected.
- Structure of schemes of finite type over a separably closed field. Varieties, the lemma stating that such a scheme is a finite disjoint union of connected and geometrically connected schemes. Supplies the geometric connectedness of the fibres of the finite type model.

Source reference. Original reference 0EZT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16410–16410. Prerequisite pointer invoked at proof line(s) 16701; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EZX: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 16610–16610. Prerequisite pointer invoked at proof line(s) 16703; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0363: sources/stacks-chow/tags.txt, source lines 4150–4150. Prerequisite pointer invoked at proof line(s) 16708; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Consider the cartesian diagrams of schemes*

$$\begin{array}{ccc} X & \xleftarrow{h} & Y \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle e} \\ S & \xleftarrow{g} & T \end{array}$$

Assume that S is the spectrum of a separably closed field. Then $f^{-1}g_*\mathcal{F} = h_*e^{-1}\mathcal{F}$ for any sheaf $\mathcal{F}$ on $T_{étale}$.

Proof. We may work locally on X. Hence we may assume X is affine. Then we can write X as a cofiltered limit of affine schemes of finite type over S. By Lemma 0EZT we may assume that X is of finite type over S. Then Lemma 0EZX applies because any scheme of finite type over a separably closed field is a finite disjoint union of connected and geometrically connected schemes (see Varieties, Lemma 0363). □